

Complete Scientific Study: Vertical Form Fill Seal (VFFS) Machines for Powder Packaging

Focus: Salt, Sugar, Coffee — 500 g packages at 15–30 bags/min

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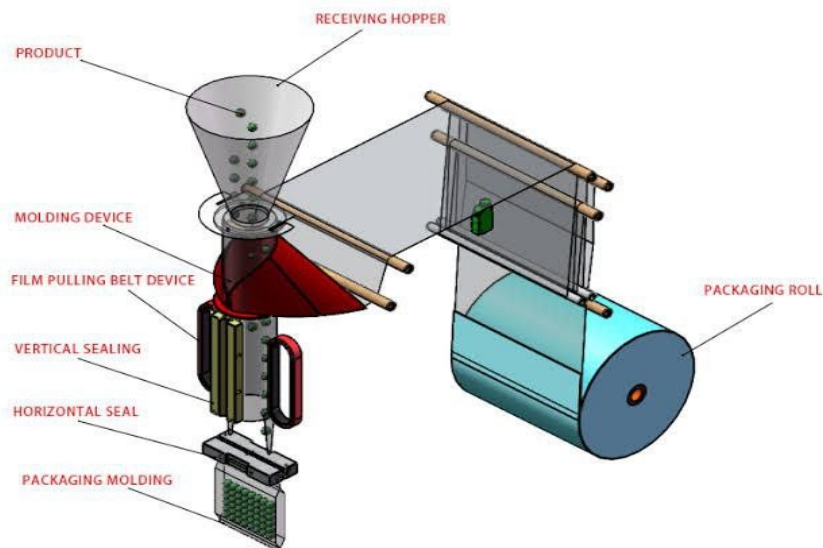
Title Page

Title: Technical and Operational Study of Vertical Form Fill Seal (VFFS) Machines for Powder Packaging — 500 g Bags at 15–30 bags/min

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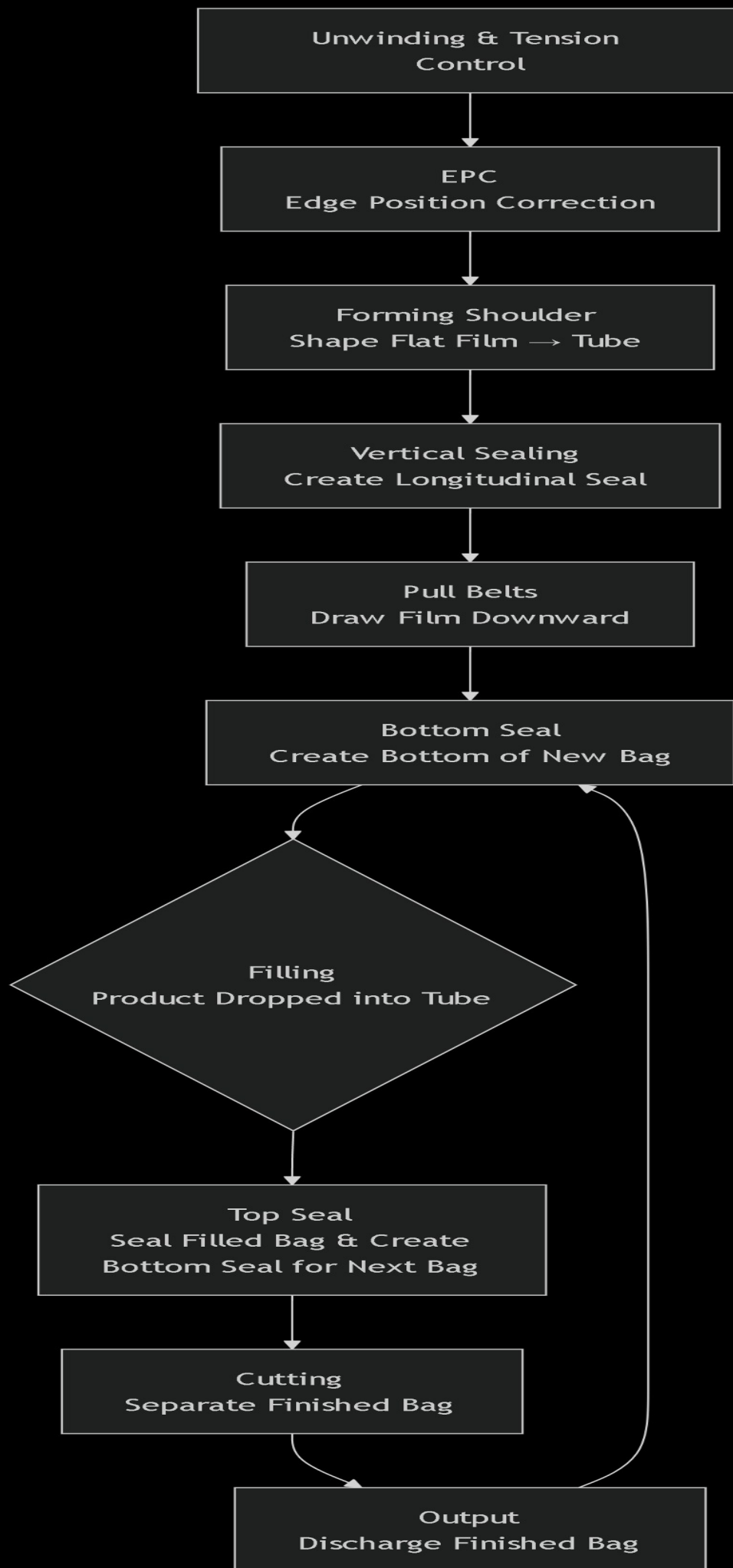


Abstract

This document consolidates a full technical study and operational guide for Vertical Form Fill Seal (VFFS) machines optimized to pack 500 g powder products — specifically salt, sugar, and coffee — at production rates between 15 and 30 bags per minute. The study preserves all supplied content and tables, refines structure and language for clarity, and adds a title page, abstract, recommendations, conclusions, diagrams, and a formatted references section. Key emphases include machine selection (intermittent vs. continuous motion), auger filler and forming collar design, film selection and film-width calculations, sealing optimization (temperature, dwell time, pressure), sensor and PLC architecture, dust control, maintenance schedules, and a complete mechanism breakdown. Practical recommendations and action items are included for immediate implementation in production.

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SECTION 1: VFFS MACHINE TYPES AND OPERATING PRINCIPLES

1.1 VFFS Machine Types Comparison

Machine Type	Operating Principle	Speed Range (bags/min)	Suitable for 15–30 bags /min	Film Stress	Mechanical Complexity	Sealing Quality	Cost	Recommended for Powders	Main Advantages
Intermittent Motion	Stop-start cycle: film stops for sealing and cutting	10–60	Yes (Optimal)	Lower	Lower	Excellent (more dwell time)	Lower	Preferred	Better seal quality, simpler design, lower cost
Continuous Motion	Continuous film movement with moving sealing jaws	60-300	Yes (Underutilized)	Higher	Higher	Good (shorter dwell time)	Higher	Optional	Higher throughput capability, less vibration

1.2 Recommendation for 500 g Powder Packaging at 15–30 bags/min

INTERMITTENT MOTION VFFS is OPTIMAL because:

- **Speed Range:** The target speed (15–30 bags/min) sits comfortably within intermittent motion machines’ optimal operating window.
- **Sealing Quality:** Intermittent motion allows longer dwell times and thus higher-quality seals — crucial to prevent dust leakage with powders.
- **Cost-Effectiveness:** Lower initial and maintenance costs for intermittent machines make them appropriate for this throughput range.

SECTION 2: COLLAR-TYPE FORMING SYSTEM AND FILM MATERIALS

2.1 Collar-Type Forming System

Component	Function	Critical Parameters
Forming Collar/Tube	Shapes flat film into tubular form around product path	Diameter: 80–120 mm for 500 g bags, Surface finish: Ra < 0.8 μm
Film Guide Rails	Guide film smoothly into collar without wrinkles	Alignment tolerance: ±0.5 mm, Material: Stainless steel 316L
Overlap Control	Controls longitudinal seal overlap (typically 8–12 mm)	Overlap precision: ±1 mm, Sensor feedback control
Tension Control	Maintains consistent film tension (2–5 N/cm width)	Tension range: 2–5 N/cm, Pneumatic or servo control

2.2 Film Materials for Powder Packaging

Film Type	Structure (μm)	Moisture Barrier	Oxygen Barrier	Heat Seal Temp (°C)	Cost Relative	Best for	Typical Thickness
PE/Paper/PE	50/40/50	Good	Poor	140–160	Low	Salt, Sugar	140
OPP/PE	20/60	Moderate	Moderate	130–150	Medium	General purpose	80
PET/PE	12/60	Excellent	Good	150–170	Medium-High	Coffee (aroma)	72
PE/AL/PE	50/9/50	Excellent	Excellent	140–160	High	Premium coffee	109

Academic Validation (preserved): Research on heat sealing of paper-based materials with PE coating shows first observable seal at **90 °C** with 2 s dwell time, while OPP/PE laminate exhibited seals at **100 °C**. (Merabteneetal., 2022 and other cited studies — see References.)

2.3 Film Width Calculations

The required film width is product-dependent because bulk density dictates package volume and thus bag dimensions.

Product	Bulk Density (kg/L)	Package Volume (L)	Bag Dimensions (cm)	Required Film Width (cm)
Salt	1.2	0.42 (417 cm³)	16.4 × 8.2 × 6.6	45.8
Sugar	0.8	0.62 (625 cm³)	18.5 × 9.3 × 7.4	49.2
Coffee	0.35	1.43 (1,429 cm³)	24.3 × 12.1 × 9.7	58.0

Notes on calculation: These film widths include allowances for longitudinal overlap, sealing margins, and trim. Confirm with actual film web lay-up and print margins when ordering film.

SECTION 3: AUGER FILLER DESIGN FOR POWDER PACKAGING

3.1 Auger Filler System Components

Component	Function	Design Specifications
Product Hopper	Stores product with consistent flow to auger	Capacity: 50–100 kg, Cone angle: 60°, Material: 316L SS
Agitator System	Prevents bridging and ensures uniform flow	Paddle/ribbon type, Speed: 10–30 RPM, Torque: 50–100 Nm
Auger Screw	Meters precise product quantity per rotation	Pitch: 0.8–1.2 × diameter, Length: 10–15 × diameter
Drive Motor	Provides variable speed control for accuracy	Servo motor, 0.5–2 kW, Encoder feedback, Speed: 1–100 RPM
Discharge Spout	Directs product into forming tube	Diameter: 40–80 mm, Anti-static coating

3.2 Auger Design Calculations

Target production: 25 bags/min of 500 g → Required feed rate: 12.5 kg/min.

Product	Bulk Density (kg/m³)	Angle of Repose (°)	Flow Characteristics	Recommended Auger Diameter (mm)	Pitch/Diameter Ratio	RPM for 500 g/bag	Volumetric Efficiency (%)
Salt	1,200	35	Free-flowing	50	1.0	15	85
Sugar	800	45	Moderate flow	60	0.9	12	80
Coffee	350	40	Poor flow	75	0.8	18	70

Technical Validation (preserved): Commercial VFFS machines with auger fillers demonstrate filling weights from 10–5,000 g with accuracy of $\leq \pm 1\%$ for powder products (supplier/industry data).

3.3 Filling Accuracy Requirements

Parameter	Salt	Sugar	Coffee
Target Weight	500 g	500 g	500 g
Tolerance (±)	5 g	7.5 g	10 g
Accuracy (%)	±1.0%	±1.5%	±2.0%
Precision (CV%)	< 0.5%	< 0.8%	< 1.2%
Calibration Frequency	Daily	Daily	Every shift

Key Design Considerations (clarified):

- **Servo motor control** with high-resolution encoder feedback improves volumetric precision and repeatability.
- **Variable pitch augers** can be used to tune volumetric delivery for low-density powders.
- **Anti-static coatings** or conductive finishes reduce powder adhesion and adhesion-related error.

SECTION 4: SEALING PROCESS OPTIMIZATION

4.1 Sealing System Components

Component	Function	Specifications
Heat Sealing Jaws	Apply heat and pressure to create seal	Material: Tool steel, PTFE coated, Width: 10–15 mm
Temperature Control	Maintains precise temperature at seal line	PID control, Accuracy: ± 2 °C, Response: < 5 s
Pressure System	Applies consistent pressure during sealing	Pneumatic/servo, Force: 500–2,000 N, Precision: $\pm 5\%$
Cooling System	Rapidly cools seal to set molecular bonds	Air/water cooling, Cooling rate: 50–100 °C/s

4.2 Temperature vs. Dwell Time Relationships

Film Type	Optimal Temp (°C)	Min Dwell Time (s)	Max Temp (°C)	Min Temp (°C)	Dwell Time Range (s)
PE/Paper/PE	150	0.8	170	130	0.5–1.5
OPP/PE	140	0.6	160	120	0.4–1.2
PET/PE	160	1.0	180	140	0.6–1.8
PE/AL/PE	150	0.8	170	130	0.5–1.5

Academic Validation (preserved): Paper-based materials with PE coating have a larger sealing window (reported up to ~220 °C in some tests) compared to many thermoplastic films which may shrink at ~140 °C. Sealing temperature and dwell time are primary drivers of seal strength; sealing pressure has a smaller influence.

4.3 Seal Quality Requirements

Parameter	Specification	Test Method	Critical for Powders
Seal Strength	≥ 15 N/15 mm	ASTM F88	Very High
Peel Strength	3–8 N/15 mm	ASTM F88	Medium
Seal Width	8–12 mm	Visual measure	High
Leak Rate	< 0.1%	Pressure decay	Very High
Visual Quality	No burns, uniform	Visual inspection	High

4.4 Sealing Optimization for 15–30 bags/min

Using a practical dwell time of 0.8 s (preserved from original content), sealing occupies a notable fraction of cycle time. Below is the seal-time fraction for common speeds:

Speed (bags/min)	Cycle Time (s)	Seal Dwell Time (s)	Seal Time as % of Cycle	Status
15	4.00	0.8	20.0%	Acceptable for single sealing station
20	3.00	0.8	26.7%	Acceptable for single sealing station
25	2.40	0.8	33.3%	Acceptable for single sealing station
30	2.00	0.8	40.0%	Acceptable for single sealing station

Engineering note: If machine cycle times are shorter than these stepwise assumptions, overlap strategies (e.g., overlapping fill and seal cycles, multiple sealing stations) or continuous-motion designs may be needed to meet higher outputs while preserving seal quality.

SECTION 5: SENSORS AND CONTROL SYSTEMS

5.1 Comprehensive Sensor Systems

Sensor Type	Technology	Location	Function	Accuracy/Range	Critical Level
Film Web Guide	Photoelectric/ Ultrasonic	Film path	Detects film misalignment and edge position	±0.1 mm	High
Temperature	RTD/Thermocouple	Sealing jaws	Monitors sealing temperature for quality control	±1 °C	Critical
Pressure	Piezoelectric	Pneumatic system	Monitors sealing pressure and pneumatic systems	±1%	Medium
Position	Linear Encoder	Auger/film feed	Tracks position of auger, film advance, cutting	±0.01 mm	Critical
Weight Check	Load Cell	Below filler	Verifies fill weight accuracy (optional)	±0.1 g	Medium
Film Tension	Strain Gauge	Film unwind	Maintains consistent film tension 2–5 N/cm	±2%	Medium
Dust Detection	Laser/Optical	Filler discharge	Detects powder dust for safety/cleaning alerts	50–500 mg/m ³	High
Bag Presence	Proximity/Photo	Bag discharge	Confirms bag presence before filling/sealing	Binary	High

5.2 Servo Control Systems

Servo-driven subsystems improve accuracy and throughput and reduce pneumatic air consumption. Example devices and recommended ratings are preserved in the following table:

Application	Motor Type	Power Rating	Encoder Resolution	Position Accuracy	Control Method	Feedback Type
Film Feed	AC Servo	0.4–1 kW	2,500 ppr	±0.1 mm	Position/Speed	Incremental
Sealing Jaw	Servo/Stepper	0.2–0.5 kW	1,000 ppr	±0.05 mm	Position	Absolute
Cutting Knife	Pneumatic/Servo	0.1–0.3 kW	500 ppr	±0.2 mm	Position	Incremental

SECTION 6: DUST CONTROL SYSTEMS

6.1 Dust Control System Components

Component	Function	Specifications	Critical for Products
Dust Extraction Unit	Removes airborne particles during filling	Flow rate: 500–1,500 m ³ /h, Vacuum: 2–5 kPa	Coffee > Sugar > Salt
HEPA Filtration	Filters air to < 0.3 µm particles (99.97% efficiency)	HEPA H13/H14, Flow: 200–800 m ³ /h, ΔP monitor	Coffee > Sugar > Salt
Ionizing Bars	Eliminates static charge buildup on film/product	AC/DC type, 7 kV, Coverage: 300 mm width	All products
Enclosure System	Contains dust within machine workspace	Polycarbonate/SS, Access panels, Interlocked	All products
Air Curtains	Creates positive pressure barrier at openings	Velocity: 5–8 m/s, Filtered air supply	Coffee > Sugar > Salt

6.2 Dust Generation Analysis by Product

Product	Particle Size Range (µm)	Dust Generation Level	Static Charge Tendency	Dust Extraction Flow (m ³ /h)	Filter Change Frequency	Health Hazard Level	Explosion Risk
Salt	100–1,000	Low	Medium	500	Monthly	Low	None
Sugar	200–800	Medium	High	800	Bi-weekly	Low	Low
Coffee	50–500	High	Very High	1,200	Weekly	Medium	Medium

6.3 Dust Control Safety Requirements

Safety Aspect	Requirement	Monitoring Method
Occupational Exposure	PEL: 10 mg/m ³ (coffee), 15 mg/m ³ (salt/sugar)	Personal samplers, Area monitors
Static Electricity	Surface resistivity: 10 ^{6–10} 12 Ω, Ionization balance: ±10 V	Static meters, Ion balance meters
Fire/Explosion	MEC: 60 g/m ³ (coffee), No ignition sources	Dust monitors, Hot work permits

SECTION 7: COMMON ISSUES AND MAINTENANCE

7.1 Common Operational Issues

Issue	Root Causes	Symptoms	Corrective Actions	Preventive Frequency
Inconsistent Fill Weight	Auger wear, bridging, servo drift, product flow variation	Weight variation > CV%, reject rate increase	Calibrate auger, check agitator, adjust speed, verify product quality	Daily check
Poor Seal Quality	Incorrect temperature, insufficient pressure, contamination, worn jaws	Weak seals, leakers, visual defects	Adjust temperature/pressure, clean jaws, replace PTFE coating	Every shift
Film Tracking Problems	Misaligned guides, uneven tension, static buildup, worn collar	Alarm activation, film breaks, crooked bags	Realign guides, adjust tension, clean ionizer, inspect collar	Daily check
Dust Leakage	Inadequate extraction, seal failure, overfilling, static charge	Powder on machine, air contamination	Check extraction system, verify seals, adjust fill volume, ionizer check	Every shift
Bag Wrinkling	Incorrect tension, misalignment, excessive speed, temperature issues	Visual defects, seal failures, bag jamming	Adjust film tension, realign forming system, reduce speed, optimize temperature	Daily check

Academic Validation (preserved): Research identified wrinkling, web buckling, and friction in forming tubes as common runnability issues with paper-based materials in VFFS machines.

7.2 Preventive Maintenance Schedule

Frequency	Tasks	Estimated Downtime	Skill Level Required	Critical for Performance
Daily	Clean auger/hopper; Check seal quality; Verify weights; Inspect film path; Clean dust filters	0.5 hr	Operator	High
Weekly	Lubricate moving parts; Check tension systems; Inspect sealing jaws; Test all sensors; Clean forming collar	2 hr	Technician	High
Monthly	Replace worn PTFE coating; Calibrate temperature sensors; Check servo encoder; Inspect pneumatic system; Filter replacement	4 hr	Technician	Medium
Quarterly	Replace sealing jaw elements; Calibrate load cells; Inspect drive chains/belts; Test safety systems; Deep clean entire machine	8 hr	Specialist	Medium
Annually	Overhaul auger system; Replace all bearings; Calibrate all sensors; Update PLC software; Professional inspection	2-3 days	Factory Service	Medium

SECTION 8: COMPLETE MECHANISM BREAKDOWN AND PROCESS FLOW

8.1 Complete VFFS Process Sequence

The VFFS process involves film being unwound from a roll, formed into a tube around a forming shoulder, vertically sealed, filled with product, then horizontally sealed and cut into individual bags.

Step	Operation	Mechanism	Time (s)	Critical Parameters
1	Film Unwinding	Motorized film unwind stand with tension control	0.1	Tension: 2–5 N/cm
2	Film Forming	Film guided through forming collar into tube shape	0.2	Alignment: ± 0.5 mm
3	Longitudinal Sealing	Vertical sealing jaw creates fin or lap seal	0.8	Temp: 130–170 °C
4	Product Filling	Auger filler dispenses measured product into tube	1.5	Accuracy: ± 1 –2%
5	Top Seal Formation	Horizontal jaw seals top of next bag	0.8	Seal: 8–12 mm width
6	Product Settling	Vibration/settling time before sealing	0.3	Vibration: 20–30 Hz
7	Bottom Seal Formation	Horizontal jaw seals bottom of current bag	0.8	Seal: 8–12 mm width
8	Cutting	Knife mechanism cuts between seals	0.2	Precision: ± 0.5 mm
9	Bag Discharge	Bag conveyor or gravity chute	0.3	Speed control
10	Quality Check	Weight check, metal detector, vision system	0.5	Weight: ± 10 g

Based on the sequence (preserved):

- **Total Cycle Time:** 5.5 s
- **Theoretical Maximum Speed:** 10.9 bags/min (If each step were strictly sequential.)
- **Practical Speed (80% efficiency):** 8.7 bags/min

Implementation note: The theoretical rate above is based on sequential timing. To reach the user’s target (15–30 bags/min) the implementation should rely on overlapping operations, parallel/separate sealing stations, or a continuous-motion architecture with moving seals. This document preserves the timing table and flags the need to re-evaluate mechanical sequencing if the target rate is mandatory.

8.2 Key Mechanisms Detailed Specifications

Mechanism	Type	Drive	Speed Range	Accuracy	Maintenance Interval
Film Pull System	Servo-driven belts	AC servo motor	10–200 m/min	± 0.1 mm	Weekly lube
Forming Collar	Conical tube	Static/fixed	N/A	± 0.5 mm	Clean daily
Vertical Sealer	Heated bar/band	Pneumatic cylinder	20–100 cycles/min	± 2 °C	Replace PTFE monthly
Horizontal Sealer	Heated jaws	Pneumatic/servo	10–60 cycles/min	± 2 °C, ± 5 N	Replace PTFE monthly
Cutting System	Rotary/reciprocating	Pneumatic/servo	10–60 cuts/min	± 0.5 mm	Sharpen quarterly

8.3 Full Component-Level Architecture of the Vertical Form-Fill-Seal (VFFS) Machine

1. Film Handling System



1.1 Film Roll Shaft

The film roll shaft serves as the primary support axis for the packaging film roll. It is typically equipped with:



- **Locking cones**, which secure the roll in a centered and stabilized position.



- **Bearing housings**, enabling low-friction rotation and ensuring smooth film unwinding during continuous operation.

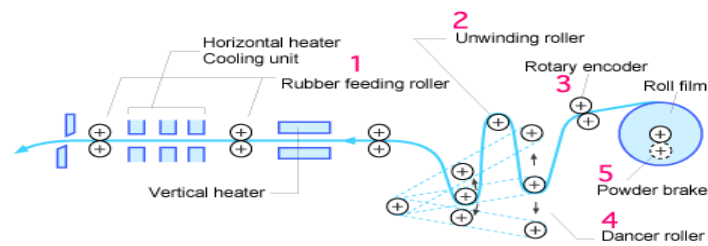


1.2 Film Unwinder Assembly

The film unwinding module regulates the release of film from the roll and ensures stable web tension. It consists of:



- **Brake mechanism** (mechanical), used to modulate resistance and prevent uncontrolled film feed.



Rubber feeding roller **1** and Unwinding roller **2** feed film.
Rotary encoder **3** and Dancer roller **4** senses the proper tension proportional to the film feed rate. Powder brake **5** increases or decreases tension based on the proper tension.

- **Tension controller**, maintaining constant film tension irrespective of roll diameter.



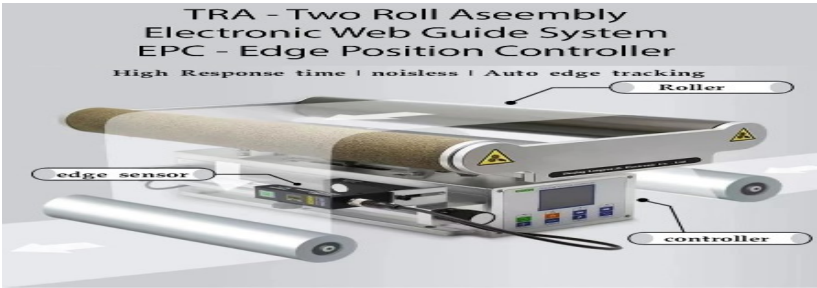
- **Dancer arm sensor**, which monitors film slack and provides real-time tension feedback.



- **Guiding rollers**, directing the film along the correct feed path and reducing lateral deviation.

1.3 Edge Position Control (EPC) System

The EPC system prevents lateral misalignment of the film during feeding.



It includes:

- **Photoelectric sensor** for detecting the film edge position.
- **Actuator** (typically servo or stepper motor) for precise corrective movements.
- **Guiding frame** that allows controlled lateral adjustments.

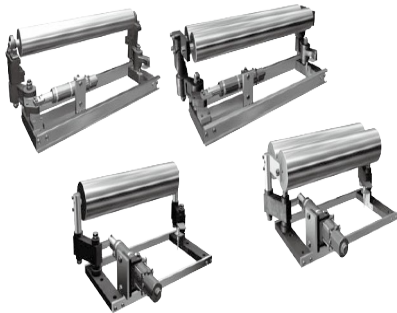
Primary function: Maintaining accurate film tracking to ensure proper forming, sealing, and printing alignment.

1.4 Guide Rollers

Guide rollers ensure uniform film travel toward the forming assembly. They are typically manufactured from:

- **Aluminum**
- **Stainless steel**
And are often coated with **anti-slip surfacing** to improve film grip. They stabilize the film immediately before entering the forming collar.

Model	Applications	Load	Roller material	Maximum tension N	Roller surface length mm
LCD-B	For business forms (has built-in width extender)	Low	Aluminum	250	550,610
LCD-01	Offset rotary press	Intermediate	Aluminum	1000	1000,1100
LCD-L	General use	Low	Aluminum or steel	250	300 ~ 650
LCD-M	General use	Intermediate	Aluminum or steel	500	500 ~ 800



2. Forming System

2.1 Forming Shoulder (Forming Collar)

The forming shoulder converts the flat film web into a continuous cylindrical tube. It is typically CNC-machined to match the film width and required package geometry. Its precision directly affects bag symmetry and seal consistency.

Name	Forming shoulders types	Size	Used and reasonings
Optimized small shoulder		100-48Co	No, because the shoulder does not have accessories for making block bottom bags. The produced pillow bags will be used to examine the surface roughness. However, the material forms without any issues.
Medium sized shoulder		140-72Co	Yes, because the fiber-based material forms smoothly over the forming shoulder with not wreckages of the material seen.
Large shoulder		180-155Co	No, because of the sharp fillet at the end of the forming shoulder and near the tube.

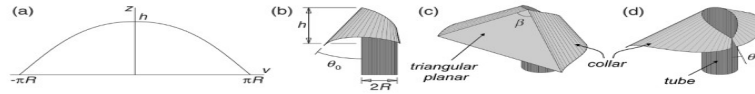


Figure 6: (a) bending curve and (b) the essential design parameters and construction of the forming shoulder. This figure was modified and reproduced from (McPherson et al., 2004).

Table 1: Common parameters based on Figure 6 that are used in the forming shoulder design (McPherson et al., 2004).

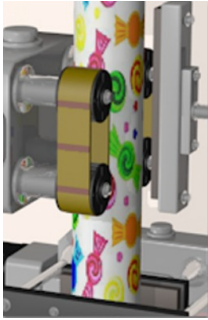
Name	Bending curve	Opening angle	Back angle	Front angle	Height	Radius
Notation	$z(v)$	β	θ_0	θ_1	h	R

Definitions

- Bending Curve**
 The variable z represents the height of a point on the bending curve. The parameter v is the circumferential coordinate around the tube, covering the arc distance from 0 to πR .
- Opening Angle (β)**
 The angle β defines the opening angle of the planar triangular region, as illustrated in Figure 6. It is also referred to as the *insertion angle*.
- Back Angle (θ_0)**
 The back angle θ_0 is defined as $\theta_0 = \theta(0)$, where θ denotes the angle between the surface normals along the bending curve at the starting point.
- Front Angle (θ_1)**
 The front angle θ_1 is defined as $\theta_1 = \theta(\pi R)$, representing the angle between the surface normals along the bending curve at the end point.
- Height (h)**
 The height h is the vertical distance between the highest and lowest points of the bending curve.
- Radius (R)**
 The parameter R denotes the radius of the tube.

2.2 Forming Tube

The forming tube supports the film as it transitions into tubular form. It may be fixed or quick-release in design.



It commonly incorporates:

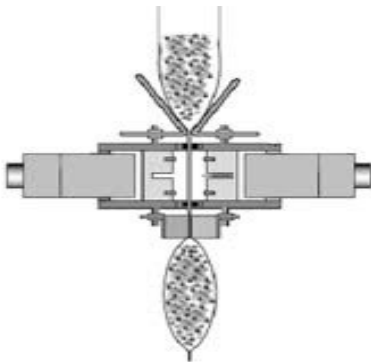
- **Tube brackets** for structural rigidity and alignment.
- **Air injection nozzle**, occasionally utilized to introduce air into the bag to maintain shape.
- **Support plates** that stabilize the forming assembly during high-speed operation.

2.3 Vertical Sealing Station (Longitudinal Seal)



This station produces the vertical (fin or lap) seal along the length of the package. It consists of:

- **Vertical sealing jaws** applying pressure and heat to create the longitudinal seal.



- **Heater cartridges** for precise thermal input.
- **PT100 temperature sensor** ensuring accurate thermal regulation.
- **Pneumatic cylinder or servo-actuated jaw** providing controlled sealing force.
- **Cooling bar** to stabilize and solidify the newly formed seal.

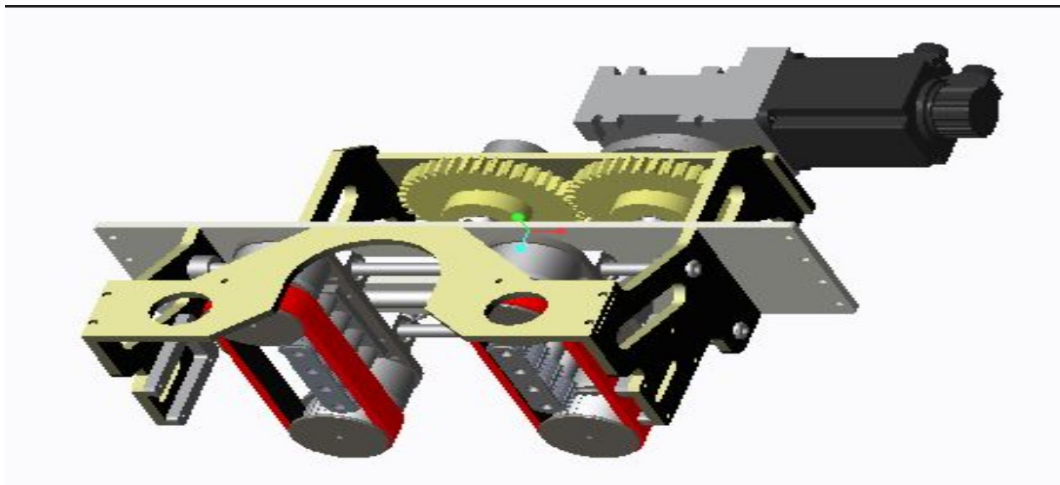
3. Film Pulling System

3.1 Pull Belts

The pull belts grip the film tube and advance it downward through the forming tube. They are typically composed of:

- **Polyurethane**
Each material type provides distinct friction, wear resistance, and thermal characteristics.

3.2 Pull Belt Assembly

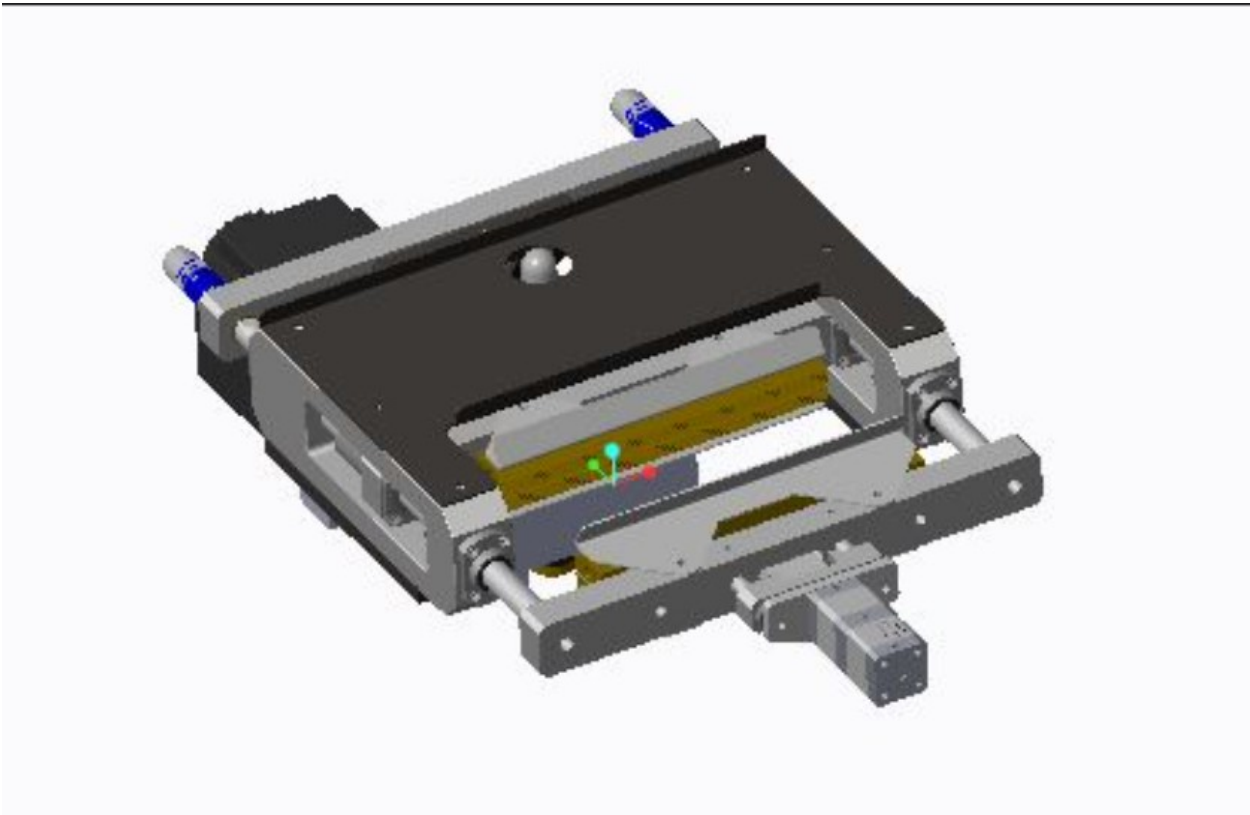


This assembly ensures accurate and consistent film feed rate. It includes:

- **Side-opening mechanism** to facilitate threading and belt maintenance.
 - **Pressure regulation unit** adjusting belt grip force for different film materials.
 - **Servo motor (pulling servo)** providing precise, programmable pulling speed.
 - **Gearbox** for torque conversion and mechanical stability.
 - **Timing belts** transmitting motion from motor to pulleys with high accuracy.
 - **Tensioner** ensuring proper belt alignment and preventing slippage.
-

4. Horizontal Sealing System

4.1 Horizontal Sealing Jaws



These jaws produce the top and bottom seals of each bag.
Two operational types exist:

- **Intermittent sealing jaws** for stop–start motion systems.
- **Rotary sealing jaws** for continuous-motion machines.

They typically include:

Table 7. VFFS pneumatic settings for different materials.



VFFS machine		Material types			
pneumatic settings		Fiber 85	Fiber 120	Plastic 15-35	Plastic 50
VFFS components	Seal jaws	6 bars	6 bars	6 bars	6 bars
	Length seal	5 bars	5 bars	5 bars	5 bars
	Belt transport	5 bars	5 bars	5 bars	6 bars
	Film brake	3 bars	3 bars	3 bars	3 bars
	Film tension	2 bars	1 bar	0.5 bars	4 bars
	Gusset	6 bars	6 bars	6 bars	6 bars

Table 6. Surface pressure calculations

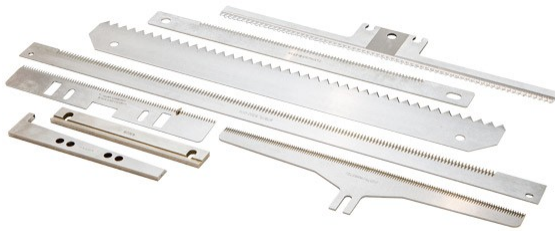
	Laboratory heat sealer	VFFS length seal	VFFS horizontal sealing tools				
Tool Name	Heat bar sealer	Longitudinal length sealer	Tool 1	Tool 2	Tool 3	Tool 4	Tool 5
Force (N)	2587,5	1256	2356	2356	2356	2356	2356
Surface area (mm ²)	6096	1920	7616	3360	2800	2240	6160
Pressure (MPa)	0.42	0.65	0.31	0.70	0.84	1.05	0.38

Table 2. Different types of sealing tool profiles.

Sealing tool profile name	Tool profile types	Seal size and description of the tool profile	Used
Tool 1		11mm Serrated profile	Yes
Tool 2		3 + 4mm Two flat surfaces	Yes
Tool 3		5mm Flat surface	Yes
Tool 4		4mm Flat surface	No
Tool 5		11mm Flat surface	No

- **Heater cartridges** delivering controlled thermal energy.
- **Thermocouple or PT100 sensors** for real-time temperature feedback.
- **Jaw bearings** enabling smooth rotational or reciprocating motion.
- **Jaw seals** ensuring airtight pressure application.
- **Jaw alignment plates** maintaining correct sealing geometry.

4.2 Cutter Knife Assembly



The cutter severs individual bags after sealing.
It may be:

- **Straight knife**
- **Zigzag knife**

4.3 Pneumatic Actuators

These actuators enable controlled mechanical motion, including:

- Movement of the horizontal sealing jaws
- Activation of the cutter knife
- Opening and closing of the pull-belt gripping system

4.4 Sensors and Control System

- a) Photo Eye Sensor



Used for detecting the printed registration mark (eye mark), ensuring precise cutting and alignment in printed films.

- b) Encoder Provides highly accurate measurement of film displacement, enabling precise bag-length control.



8.4 Engineering Calculations for Actuation, Motion Control, and Pneumatic Sizing in VFFS Machines

Salt Filling System and Stepper Motor Calculations

Salt Filling System Calculations

Number of Revolutions per Bag

The required number of screw revolutions to fill a salt bag of mass M_{bag} is given by

$$N = \frac{M_{bag}}{m_{rev}},$$

where m_{rev} is the mass of salt delivered per revolution of the auger or filling screw. For example, if one revolution dispenses 0.025 kg of salt, then a 100 g bag (0.100 kg) requires

$$N = \frac{0.100}{0.025} = 4$$

revolutions. Similarly, a 300 g bag requires $N = 12$ revolutions and a 500 g bag requires $N = 20$ revolutions under the same assumption.

Filling Time per Bag

If the screw (or motor) rotates at n revolutions per minute (RPM), the time T_{fill} to fill one bag is

$$T_{\text{fill}} = \frac{N}{n} \times 60 \text{ seconds},$$

since each revolution takes $60/n$ seconds. For instance, using $n = 95.5$ RPM (see below) and $N = 4$ revolutions for a 100 g bag,

$$T_{\text{fill}} = \frac{4}{95.5} \times 60 \approx 2.51 \text{ s}.$$

Similarly, for $N = 12$ (300 g) and $N = 20$ (500 g), $T_{\text{fill}} \approx 7.53$ s and 12.56 s, respectively, at the same speed.

Motor Speed (RPM) and Angular Velocity

The motor speed in RPM is determined by the drive pulse rate and step angle (see Stepper section). Once the steady speed n (RPM) is set, the angular velocity ω in rad/s is

$$\omega = \frac{2\pi n}{60}.$$

For example, $n = 95.5$ RPM corresponds to $\omega \approx 2\pi \times 95.5 / 60 \approx 10.0$ rad/s.

Bag Throughput (Bags per Minute)

The throughput R in bags per minute is the reciprocal of the filling time per bag. If each bag takes T_{fill} seconds, then

$$R = \frac{60}{T_{\text{fill}}}.$$

Thus at $T_{\text{fill}} \approx 2.51$ s (100 g bag above), $R \approx 60 / 2.51 \approx 23.9$ bags/min. For $T_{\text{fill}} \approx 7.53$ s (300 g), $R \approx 7.97$ bags/min; for 12.56 s (500 g), $R \approx 4.78$ bags/min.

Hopper Capacity

The hopper must hold sufficient salt for the desired production run. If each bag has mass M_{bag} , the production rate is $M_{\text{bag}} \times R$ (kg per minute). For a continuous run of T_{run} minutes, the required hopper mass capacity C_{mass} is

$$C_{\text{mass}} = M_{\text{bag}} \times R \times T_{\text{run}}.$$

For example, at 0.1 kg per bag, 10 bags/min, and 60 minutes, $C_{\text{mass}} = 0.1 \times 10 \times 60 = 60$ kg of salt. To convert to volume, divide by the bulk density of salt (around 1200 kg/m³): e.g. 60 kg is about 0.05 m³ (50 liters).

Stepper Motor Calculations for the Feed System

Number of Pulses

Let l be the required linear feed distance (e.g. for the auger or dispenser) and let $l_{\text{rev}} = \pi D$ be the travel per revolution of the drive pulley of diameter D . If the stepper motor has step angle θ_s (degrees), the total number of pulses A needed for distance l is

$$A = \frac{l}{l_{\text{rev}}} \cdot \frac{360^\circ}{\theta_s}.$$

For example, with $l = 0.150$ m, $D = 0.020$ m (so $l_{\text{rev}} = 0.0628$ m), and $\theta_s = 1.8^\circ$, one finds

$$A = \frac{0.150}{0.0628} \cdot \frac{360}{1.8} \approx 478 \text{ pulses}.$$

Feed Distance and Movement per Pulse

The linear movement per pulse is

$$\Delta l_{\text{per pulse}} = \frac{l_{\text{rev}} \theta_s}{360^\circ}.$$

Using $l_{\text{rev}} = 0.0628$ m and $\theta_s = 1.8^\circ$, this gives $\Delta l_{\text{per pulse}} \approx 0.000314$ m (0.314 mm) per step.

Required Pulse Frequency

If A pulses must be sent in a positioning time t_0 , the average pulse frequency f in Hz is A/t_0 . Including acceleration, a general formula is

$$f = \frac{A - f_1 t_1}{t_0 - t_1},$$

where f_1 is the initial pulse rate (Hz) and t_1 the accel/decel interval. For a simple case with no acceleration ramp ($f_1 = 0$) and $t_0 = 1.5$ s, one gets $f = 478 / 1.5 \approx 318.7$ Hz.

The motor speed in RPM is then

$$n = \frac{f \theta_s}{360^\circ} \times 60.$$

With $f \approx 318.5$ Hz and $\theta_s = 1.8^\circ$, this yields $n \approx 95.5$ RPM (consistent with the example above).

Angular Acceleration

The angular acceleration α (rad/s²) when increasing speed from 0 to final ω_f over time t_1 is

$$\alpha = \frac{\omega_f}{t_1},$$

with $\omega_f = 2\pi n / 60$ rad/s. Using $n = 95.5$ RPM ($\omega_f \approx 10.0$ rad/s) and $t_1 = 0.5$ s, one finds $\alpha \approx 20$ rad/s².

Load Torque

The load torque T_L arises from the weight and friction of the feed mechanism. If the effective load force is F_L acting at radius r , then

$$T_L = F_L r.$$

For example, if $F_L = 4.24$ N (including friction) and $r = 0.30$ m, then $T_L = 4.24 \times 0.30 = 1.27$ N·m.

Acceleration Torque

The torque needed to accelerate the rotor and load is

$$T_a = (J_0 i^2 + J_L) \alpha,$$

where J_0 is the motor rotor inertia, J_L the reflected load inertia, and i the gear ratio. Using $\alpha \approx 20$ rad/s² and typical inertias (e.g. $J_L \approx 2.25 \times 10^{-4}$ kg·m² for a drum of 0.06 m diameter and 0.5 kg load), one computes T_a . In many designs T_a is on the order of 0.01–0.1 N·m.

Total Required Torque

The total required motor torque T_{required} includes both load and acceleration torques with a safety factor S_f :

$$T_{\text{required}} = (T_L + T_a) S_f.$$

For example, with $T_L = 1.27$ N·m, $T_a \approx 0.01$ N·m, and $S_f = 2$,

$$T_{\text{required}} \approx (1.27 + 0.01) \times 2 \approx 2.56 \text{ N·m}.$$

Each of the above formulas can be included in the report with appropriate parameter values for the specific machine design. All equations are presented in standard form suitable for technical documentation.

8.5 Power and Utilities Requirements

Commercial VFFS machines typically require 2.2–5.6 kW power and compressed air at 0.5–0.8 MPa with consumption of 0.3–0.5 m³/min.

Utility	Specification	Consumption	Connection Size	Critical for
Electrical Power	1-phase, 220 V, 50/60 Hz	1–4 kW total	32 A circuit breaker	All operations
Compressed Air	6–8 bar, Dry, Oil-free	50–100 L/min (NL)	1/2" NPT	Sealing, cutting, controls
Cooling Water	15–25 °C, 2–4 L/min	120–240 L/hr	3/4" hose	Seal cooling (optional)
Exhaust Ventilation	500–1,500 m ³ /h	Product dependent	150–200 mm duct	Dust control, worker safety

SECTION 9: REFERENCES AND KEYWORDS FOR ACADEMIC RESEARCH

9.1 Recommended Academic Databases

Database	Focus Area	Relevance Score
Packaging Technology & Science	Packaging-specific research	Very High
IEEE Xplore	Control systems, automation, sensors	High
ScienceDirect	Materials science, food processing	High
Web of Science	Multidisciplinary engineering research	High
Food and Bioproducts Processing	Food packaging applications	High

9.2 Key Research Areas for Further Investigation

- Auger filler design: numerical simulation (CFD) and experimental validation.
- Process control: closed-loop control algorithms to improve fill accuracy.
- Predictive maintenance: sensor fusion and ML models for component failure forecasting.
- Sustainable packaging: evaluation of biodegradable films compatible with VFFS.

SECTION 10: ACADEMIC AND TECHNICAL REFERENCES

10.1 Peer-Reviewed Research Cited

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10.2 Technical Standards Cited

4. ASTM F88/F88M-23: "Standard Test Method for Seal Strength of Flexible Barrier Materials."

10.3 Industry and Commercial Sources

11. Spack Machine (2024): "VFFS Machines Guide: Working Principle, Types, Filling Methods"
 12. Boshi Pack: "CE 500 g automatic VFFS baking powder packaging machine"
 13. Alibaba Supplier: "VFFS 500 g-5 kg Powder Packaging Machine Specifications"
 14. Chantec Pack (2025): "Vertical Form Fill Seal Machines vs. Traditional Packaging Methods"
 15. Packaim (2024): "The Complete Guide to VFFS Machines"
 16. Alibaba Supplier: "VFFS 500 g-5 kg Powder Packaging Machine Specifications" (Used for power and air data).
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Conclusions & Recommendations

Conclusions

- Intermittent-motion VFFS machines are recommended for the target production range (15–30 bags/min) because they achieve excellent seal quality and are cost-effective for this throughput.
- Achieving precise volumetric fill for low-density materials (coffee) requires larger augers, variable pitch geometry, and close servo control. Calibration frequency must increase with poorer-flow powders.
- Dust control is critical for coffee and sugar: provide HEPA filtration, strong extraction, ionization, and a sealed enclosure to maintain worker safety and product quality.
- The sequential timing in Section 8 shows that reaching 15–30 bags/min requires overlapping cycles or continuous motion; a process rework is suggested if the user must meet the higher end of the range.

Practical Recommendations (Actionable)

1. **Machine selection:** Procure an intermittent-motion VFFS with servo-driven film feed and at least one auger station. Consider dual horizontal sealing stations if aiming >25 bags/min.
 2. **Film specification:** Use PET/PE or PE/AL/PE for coffee to preserve aroma; PE/paper/PE for salt and sugar to reduce cost.
 3. **Dust control:** Install a dedicated dust extraction unit ($\geq 1,200 \text{ m}^3/\text{h}$ for coffee lines) with HEPA filtration and ionizing bars.
 4. **Control and sensing:** Integrate PLC + HMI, high-resolution encoders, and in-line weight-checking load cells for closed-loop verification.
 5. **Maintenance:** Implement the preventive schedule in Section 7 and log maintenance events in an electronic CMMS.
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